Algorithm 2 Alternating Optimization to solve (10)

Initialize $i \leftarrow 0, \boldsymbol{\Theta}_0, \mathbf{W}_0, \eta_0$.
repeat
  Update $\mathbf{W}^{(i+1)}$ using Algorithm 1.
  Optimize $\boldsymbol{\Theta}$ via manifold-based optimization:
    while no convergence of $\|\nabla_{\mathcal{M}} f_2(\boldsymbol{\Theta}_i)\|_F$
      $i \leftarrow i+1$;
      Calculate δ_i using backtracking algorithms [39];
      Update $\boldsymbol{\Theta}_i$ using (38).
      Update κ_i using (37).
      Update ξ_i using (36).
    end while
  Obtain $\boldsymbol{\Theta}^\star$ by enforcing BD-RIS constraints:
    if $G = 1$ (Fully-Connected)
      $\boldsymbol{\Theta}^\star_{\mathrm{FC}} = \mathsf{symuni}(\boldsymbol{\Theta}_{i+1})$.
    elseif $G = N$ (Single-Connected)
      $\boldsymbol{\Theta}^\star_{\mathrm{SC}} = \mathsf{diag}(e^{1\mathrm{j}\angle x_1}, \ldots, e^{1\mathrm{j}\angle x_N})$.
    else (Group-Connected)
      Block-diagonalize $\boldsymbol{\Theta}_{i+1}$ using (40).
      $\boldsymbol{\Theta}^\star_{\mathrm{GC}} = \mathsf{diag}\left(\mathsf{symuni}(\mathbf{X}_1), \ldots, \mathsf{symuni}(\mathbf{X}_G)\right)$.
until $\left|\mathrm{WSR}(t) - \mathrm{WSR}(t-1)\right| \leq \epsilon$.
Return $\mathbf{W}^\star, \boldsymbol{\Theta}^\star$.